JEE Main Online Exam 2024

Questions & Solution

09th April 2024 | Morning

PHYSICS

Section-A: This section contains 20 multiple choice questions. Each question has 4 choices(1), (2), (3) and (4), out of which **ONLY ONE** is correct..

Q.1 A proton, an electron and an alpha particle have the same energies. Their de-Broglie wavelengths will be compared as: a

(1) $\lambda_e > \lambda_\alpha > \lambda_p$ (2) $\lambda_\alpha < \lambda_p < \lambda_e$ (3) $\lambda_p < \lambda_e < \lambda_\alpha$ (4) $\lambda_p > \lambda_e > \lambda_\alpha$

Ans. **[2]**

Sol. $\lambda_{DB} = \frac{h}{p} = \frac{h}{\sqrt{2mk}}$

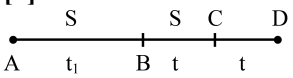
$\Rightarrow \lambda_{DB} \alpha \frac{1}{\sqrt{m}} \Rightarrow \lambda_\alpha < \lambda_p < \lambda_e$

Q.2 A particle moving in a straight line covers half the distance with speed 6 m/s. The other half is covered in two equal time intervals with speeds 9 m/s and 15 m/s respectively. The average speed of the particle during the motion is :

(1) 8.8 m/s (2) 10 m/s (3) 9.2 m/s (4) 8 m/s

Ans. **[4]**

Sol.



$BD \Rightarrow S = 9t + 15t = 24t$

$AB \Rightarrow S = 6t_1 = 24t \Rightarrow t_1 = 4t$

$< \text{speed} > = \frac{\text{dist.}}{\text{time}} = \frac{48t}{2t + t_1}$

$= \frac{48t}{2t + 4t_1} \Rightarrow \frac{48t}{6t} \Rightarrow 8 \text{ m/s}$

Q.3 A plane EM wave is propagating along x direction. It has a wavelength of 4 mm. If electric field is in y direction with the maximum magnitude of 60 Vm^{-1}, the equation for magnetic field is:

(1) $B_z = 60 \sin\left[\frac{\pi}{2}(x - 3\times10^8 t)\right]\hat{k}\,T$ (2) $B_z = 2 \times 10^{-7} \sin\left[\frac{\pi}{2}\times10^3(x - 3\times10^8 t)\right]\hat{k}\,T$

(3) $B_x = 60\sin\left[\frac{\pi}{2}(x - 3\times10^8 t)\right]\hat{i}\,T$ (4) $B_z = 2 \times 10^{-7} \sin\left[\frac{\pi}{2}(x - 3\times10^8 t)\right]\hat{k}\,T$

Ans. **[2]**

Sol. $E = BC \Rightarrow 60 = B \times 3 \times 10^8$

$\Rightarrow B = 2 \times 10^{-7}$

Also $C = f\lambda$

$\Rightarrow 3 \times 10^8 = f \times 4 \times 10^{-3}$

$\Rightarrow f = \frac{3}{4} \times 10^{11}$

$\Rightarrow \omega = 2\pi f = \frac{3}{4} \times 2\pi \times 10^{11}$

$\Rightarrow \omega = \frac{\pi}{2} \times 10^{3} C$

$\Rightarrow$ Electric field $\Rightarrow$ y direction

Propagation $\Rightarrow$ x direction

Magnetic field $\Rightarrow$ z-direction

Q.4 Given below are two statements:

Statement (I): When an object is placed at the centre of curvature of a concave lens, image is formed at the centre of curvature of the lens on the other side.

Statement (II): Concave lens always forms a virtual and erect image.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) **Statement I** is false but **Statement II** is true. (2) Both **Statement I** and **Statement II** are false.

(3) **Statement I** is true but **Statement II** is false. (4) Both **Statement I** and **Statement II** are true.

Ans. **[1]**

Sol. Option 1 will be correct if we also consider virtual objects

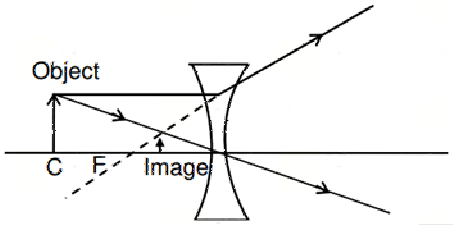


Image $\rightarrow$ erect, virtual, same side of lens

Statement (I) is incorrect

Statement (II) is correct if we cosider only real objects, if we also consider virtual objets then statement-(II) will be incorrect.

For u = f/2 (virtual object)

$v = \frac{u.f}{u+f}$

$v = \frac{(f/2)(-f)}{(f/2)+(-f)} = -\frac{f^2}{-f}$

$v = f \Rightarrow$ real image

Q.5 A light emitting diode (LED) is fabricated using GaAs semiconducting material whose band gap is 1.42 eV. The wavelength of light emitted from the LED is:

(1) 650 nm (2) 1243 nm (3) 875 nm (4) 1400 nm

Ans. **[3]**

Sol. $\lambda = \frac{1240}{1.42} = 875$ nm (Approx)

Q.6 A sphere of relative density σ and diameter D has concentric cavity of diameter d. The ratio of D/d, if it just floats on water in a tank is:

(1) $\left(\frac{\sigma}{\sigma-1}\right)^{1/3}$ (2) $\left(\frac{\sigma+1}{\sigma-1}\right)^{1/3}$ (3) $\left(\frac{\sigma-1}{\sigma}\right)^{1/3}$ (4) $\left(\frac{\sigma-2}{\sigma+1}\right)^{1/3}$

Ans. **[1]**

Sol. $\text{weitht (w)} = \frac{4}{3}\pi\left(\frac{D^3 - d^3}{8}\right)\sigma g$

$\text{Buoyant force } (F_b) = 1 \times \frac{4}{3}\pi\left(\frac{D^3}{8}\right).g$

For Just Float $\Rightarrow w = F_b$

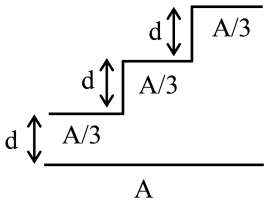
$\Rightarrow (D^3 - d^3)\,\sigma = D^3$

$\Rightarrow 1 - \frac{d^3}{D^3} = \frac{1}{\sigma}$

$\Rightarrow 1 - \frac{1}{\sigma} = \left(\frac{d}{D}\right)^3$

$\Rightarrow \left(\frac{\sigma}{\sigma - 1}\right)^{1/3} = \left(\frac{D}{d}\right)$

Q.7 A capacitor is made of a flat plate of area A and a second plate having a stair-like structure as shown in figure. If the area of each stair is A/3 and the height is d, the capacitance of the arrangement is:



(1) $\frac{11\varepsilon_0 A}{18d}$ (2) $\frac{13\varepsilon_0 A}{17d}$ (3) $\frac{11\varepsilon_0 A}{20d}$ (4) $\frac{18\varepsilon_0 A}{11d}$

Ans. **[1]**

Sol. All capacitor are in parallel combination.
Also effective area is common area only

$\Rightarrow C_{eq} = C_1 + C_2 + C_3$

$\Rightarrow C_{eq} = \frac{A\varepsilon_0}{3d} + \frac{A\varepsilon_0}{3(2d)} + \frac{A\varepsilon_0}{3(3d)}$

$\Rightarrow C_{eq} = \frac{A\varepsilon_0}{3}\left(\frac{11}{6d}\right)$

$\Rightarrow C_{eq} = \frac{11A\varepsilon_0}{18d}$

Q.8 A light unstretchable string passing over a smooth light pulley connects two blocks of masses m_1 and m_2. If the acceleration of the system is g/8, then the ratio of the masses m_2/m_1 is:
(1) 9 : 7 (2) 4 : 3 (3) 5 : 3 (4) 8 : 1

Ans. **[1]**

Sol. $a_{sys} = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)g = \frac{g}{8}$

$\Rightarrow \frac{m_2}{m_1} = \frac{9}{7}$

Q.9 The dimensional formula of latent heat is:

(1) $[M^0LT^{-2}]$ (2) $[MLT^{-2}]$ (3) $[M^0L^2T^{-2}]$ (4) $[ML^2T^{-2}]$

Ans. **[3]**

Sol. Latent heat is specific heat

$$\Rightarrow \frac{ML^2T^{-2}}{M} = M^0L^2T^{-2}$$

Q.10 The volume of an ideal gas ($\gamma = 1.5$) is changed adiabatically from 5 litres to 4 litres. The ratio of initial pressure to final pressure is:

(1) $\frac{4}{5}$ (2) $\frac{16}{25}$ (3) $\frac{8}{5\sqrt{5}}$ (4) $\frac{2}{\sqrt{5}}$

Ans. **[3]**

Sol. For Adiabatic process

$P_iV_i^{\gamma} = P_fV_f^{\gamma}$

$P_i(5)^{1.5} = P_f(4)^{1.5}$

$$\frac{P_i}{P_f} = \left(\frac{4}{5}\right)^{3/2} = \frac{4}{5}.\left(\frac{4}{5}\right)^{1/2} \Rightarrow \frac{8}{5\sqrt{5}}$$

Q.11 The energy equivalent of 1g of substance is:

(1) 11.2×10^{24} MeV (2) 5.6×10^{12} MeV (3) 5.6 eV (4) 5.6×10^{26} MeV

Ans. **[4]**

Sol. $E = mC^2$

$\Rightarrow E = (1 \times 10^{-3}) \times (3 \times 10^8)^2 J$

$$E = \frac{9\times10^{13}}{1.6\times10^{-19}} e.V.$$

$E = 5.6 \times 10^{26}$ MeV

Q.12 An astronaut takes a ball of mass m from earth to space. He throws the ball into a circular orbit about earth at an altitude of 318.5 km. From earth's surface to the orbit, the change in total mechanical energy of the ball is $x\frac{GM_em}{21R_e}$. The value of x is (take $R_e = 6370$ km):

(1) 11 (2) 9 (3) 12 (4) 10

Ans. **[1]**

Sol. $h = 318.5 \approx \left(\frac{R_e}{20}\right)$

$$T.E_i = \frac{-GM_em}{R_e}$$

$$T.E_f = \frac{-GM_em}{2(R_e + h)} = \frac{-GM_em}{2\left(R_e + \frac{R_e}{20}\right)}$$

$$\Rightarrow T.E_f = \frac{-10GM_em}{21R_e}$$

Change in total mechanical energy

$= TE_f - TE_i$

$$= \frac{GM_em}{R_e}\left[1 - \frac{10}{21}\right] = \frac{11GM_em}{21R_e}$$

Q.13 Given below are two statements:
Statement (I) : When currents vary with time, Newton's third law is valid only if momentum carried by the electromagnetic field is taken into account.
Statement (II) : Ampere's circuital law does not depend on Biot-Savart's law.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both **Statement I** and **Statement II** are false.
(2) **Statement I** is true but **Statement II** is false.
(3) **Statement I** is false but **Statement II** is true.
(4) Both **Statement I** and **Statement II** are true.

Ans. **[2]**

Sol. Conceptual

Q.14 A particle of mass m moves on a straight line with its velocity increasing with distance according to the equation $v = \alpha\sqrt{x}$, where α is a constant. The total work done by all the forces applied on the particle during its displacement from x = 0 to x = d, will be:

(1) $\frac{m}{2\alpha^2 d}$ (2) $\frac{md}{2\alpha^2}$ (3) $\frac{m\alpha^2 d}{2}$ (4) $2m\alpha^2 d$

Ans. **[3]**

Sol. $v = \alpha\sqrt{x}$

at = x = 0; v = 0

& at x = d; $v = \alpha\sqrt{d}$

$W.D = K_f - K_i$

$W.D = \frac{1}{2}m(\alpha\sqrt{d})^2 - \frac{1}{2}m(0)^2$

$\Rightarrow W.D = \frac{m\alpha^2 d}{2}$

Q.15 A galvanmeter has a coil of resistance 200 Ω with a full scale deflection at 20 μA. The value of resistance to be added to use it as an ammeter of range (0 – 20) mA is:

(1) 0.40 Ω (2) 0.20 Ω (3) 0.50 Ω (4) 0.10 Ω

Ans. **[2]**

Sol. $i_g = 20\ \mu A$

$i = i_g\left(\frac{G}{S}+1\right)$

$\Rightarrow 20\times10^{-3} = 20\times10^{-6}\left(\frac{200}{S}+1\right)$

$\Rightarrow \frac{200}{S} = 999$

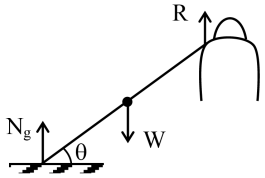
$\Rightarrow S \approx 0.2\Omega$

Q.16 A heavy iron bar, of weight W is having its one end on the ground and the other on the shoulder of a person. The bar makes an angle θ with the horizontal. The weight experienced by the person is:

(1) W/2 (2) W (3) W cos θ (4) W sin θ

Ans. **[1]**

Sol.



R = net reaction force by shoulder

Balancing torque about pt of contact on ground :

$$W\left(\frac{L}{2}\cos\theta\right) = R(L\cos\theta)$$

$$\Rightarrow R = \frac{W}{2}$$

Q.17 One main scale division of a vernier caliper is equal to m units. If nth division of main scale coincides with $(n+1)^{th}$ division of vernier scale, the least count of the vernier caliper is:

(1) $\frac{n}{(n+1)}$ (2) $\frac{m}{(n+1)}$ (3) $\frac{1}{(n+1)}$ (4) $\frac{m}{n(n+1)}$

Ans. **[2]**

Sol. $nMSD = (n+1)\,VSD$

$$\Rightarrow 1\,VSD = \frac{n}{n+1}MSD$$

$$L.C = 1\,MSD - 1\,VSD$$

$$L.C = m - m\left(\frac{n}{n+1}\right)$$

$$L.C = m\left(\frac{n+1-n}{n+1}\right)$$

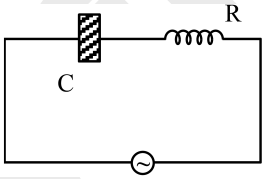
$$\Rightarrow L.C = \left(\frac{m}{n+1}\right)$$

Q.18 A bulb and a capacitor are connected in series across an ac supply. A dielectric is then placed between the plates of the capacitor. The glow of the bulb:

(1) increases (2) remains same (3) becomes zero (4) decreases

Ans. **[1]**

Sol.



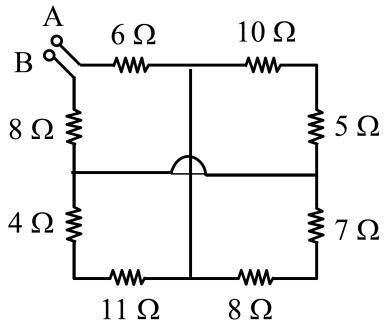
$$Z = \sqrt{R^2 + X_C^2} \;\&\; X_C = \frac{1}{\omega C}$$

due to dielectric

$C\uparrow \Rightarrow X_C\downarrow \Rightarrow Z\downarrow$

So, current increases & thus bulb will glow more brighter.

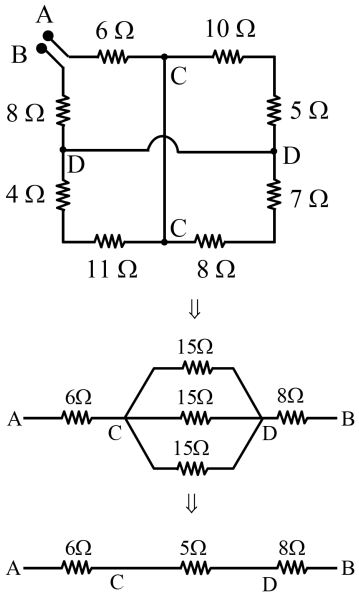
Q.19 The equivalent resistance between A and B is:



(1) 18 Ω (2) 25 Ω (3) 27 Ω (4) 19 Ω

Ans. **[4]**

Sol.



$R_{eq} = 6\Omega + 5\Omega + 8\Omega = 19\Omega$

Q.20 A sample of 1 mole gas at temperature T is adiabatically expanded to double its volume. If adiabatic constant for the gas is $\gamma = 3/2$, then the work done by the gas in the process is :

(1) $RT[2-\sqrt{2}]$ (2) $\frac{R}{T}[2-\sqrt{2}]$ (3) $RT[2+\sqrt{2}]$ (4) $\frac{T}{R}[2+\sqrt{2}]$

Ans. **[1]**

Sol. $TV^{\gamma-1} = \text{constant}$

$\Rightarrow T(V)^{\frac{3}{2}-1} = T_f(2V)^{\frac{3}{2}-1}$

$\Rightarrow TV^{1/2} = T_f(2)^{1/2}(V)^{1/2}$

$\Rightarrow T_f = \left(\frac{T}{\sqrt{2}}\right)$

Now, $W.D. = \frac{nR\Delta T}{1-\gamma} = \frac{1.R\left[\frac{T}{\sqrt{2}} - T\right]}{1-\frac{3}{2}}$

$\Rightarrow$ W.D. $= 2RT\left[1-\frac{1}{\sqrt{2}}\right]$

$\Rightarrow$ W.D $= RT[2-\sqrt{2}\,]$

Section-B: Numerical Value Type Questions: This section contains 10 Numerical based questions. Attempt any 5 questions out of 10. The answer to each question should be rounded-off to the nearest integer.

Q.21 If $\vec{a}$ and $\vec{b}$ makes an angle $\cos^{-1}\left(\frac{5}{9}\right)$ with each other, then $|\vec{a}+\vec{b}| = \sqrt{2}\,|\vec{a}-\vec{b}|$ for $|\vec{a}| = n|\vec{b}|$. The integer value of n is .....

Ans. **[3]**

Sol. $\cos\theta = \frac{5}{9}$

$\frac{\vec{a}.\vec{b}}{ab} = \frac{5}{9}$ ....(1)

$|\vec{a}+\vec{b}| = \sqrt{2}\,|\vec{a}-\vec{b}|$

$a^2 + b^2 + 2\vec{a}.\vec{b} = 2a^2 + 2b^2 - 4\vec{a}.\vec{b}$

$6\vec{a}.\vec{b} = a^2 + b^2$

$6 \times \frac{5}{9}ab = a^2 + b^2$

$\frac{10}{3}ab = a^2 + b^2$ & $a = nb$

$\frac{10}{3}nb^2 = n^2b^2 + b^2$

$3n^2 - 10n + 3 = 0$

$n = \frac{1}{3}$ and $n = 3$

integer value $n = 3$

Q.22 At the centre of a half ring of radius R = 10 cm and linear charge density 4n C m^{-1}, the potential is x π V. The value of x is _____.

Ans. **[36]**

Sol. Potential at centre of half ring

$V = \frac{KQ}{R}$

$V = \frac{K\lambda\pi R}{R}$

$V = K\lambda\pi \Rightarrow V = 9 \times 10^9 \times 10^{-9}\pi$

$V = 36\pi$

Q.23 A star has 100% helium composition. It starts to convert three ^4He into one ^{12}C via triple alpha process as $^4He + {}^4He + {}^4He \rightarrow {}^{12}C + Q$. The mass of the star is 2.0×10^{32} kg and it generates energy at the rate of 5.808×10^{30} W. The rate of converting these ^4He to ^{12}C is $n \times 10^{42}$ s^{-1}, where n is _____.
[Take, mass of ^4He = 4.0026 u, mass of ^{12}C = 12 u]

Ans. **[15]** **NTA Ans. [5]**

Sol. $^4He + {}^4HE + {}^4HE \rightarrow {}^{12}C + Q$

power generated = $\frac{N}{t}Q$

where, N $\rightarrow$ No. of reaction/sec.

$Q = (3m_{He} - m_C)C^2$

$Q = (3 \times 4.0026 - 12)(3 \times 10^8)^2$

$Q = 7.266$ MeV

$$\frac{N}{t} = \frac{\text{power}}{Q} = \frac{5.808 \times 10^{30}}{7.266 \times 10^6 \times 1.6 \times 10^{-19}}$$

$$\frac{N}{t} = 5 \times 10^{42}$$

rate of conversion of ^4He into $^{12}C = 15 \times 10^{30}$

Hence, n = 15

Q.24 In a Young's double slit experiment, the intensity at a point is $\left(\frac{1}{4}\right)^{th}$ of the maximum intensity, the minimum distance of the point from the central maximum is ..... μm. (Given : λ = 600 nm, d = 1.0 nm, D = 1.0 m)

Ans. **[200]**

Sol. $I = I_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$

$$\frac{I_0}{4} = I_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$$

$$\Delta\phi = \frac{2\pi}{3}$$

$$\frac{2\pi}{\lambda}\left(\frac{yd}{D}\right) = \frac{2\pi}{3}$$

$$y = \frac{\lambda D}{3d} = \frac{600 \times 10^{-19} \times 1}{3 \times 10^{-3}} = 2 \times 10^{-4} \text{ m}$$

Q.25 A string is wrapped around the rim of a wheel of moment of inertia 0.40 kgm^2 and radius 10 cm. The wheel is free to rotate about its axis. Initially the wheel is at rest. The string is now pulled by a force of 40 N. The angular velocity of the wheel after 10s is x rad/s, where x is _____.

Ans. **[100]**

Sol. $\tau = FR = I\alpha \Rightarrow 40 \times 0.1 = 0.4\alpha$

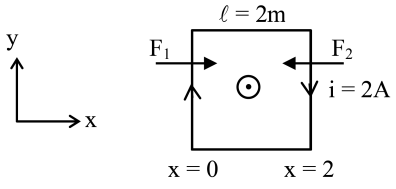
$\alpha = 10$ rad/s^2

$W_f = 10 \times 10 = 100$ rad/s

Q.26 A square loop of edge length 2 m carrying current of 2 A is placed with its edges parallel to the x-y axis. A magnetic field is passing through the x-y plane and expressed as $\vec{B} = B_0(1 + 4x)\hat{k}$, where B_0 = 5 T. The net magnetic force experienced by the loop is _____ N.

Ans. **[160]**

Sol.



$B(x = 0) = B_0$, $B(x = 2) = 9B_0$

Also, $F = i\ell B$

$\Rightarrow F_1 = i\ell B_0$ & $F_2 = 9i\ell B_0$

$F = F_2 - F_1 = 8i\ell B_0 = 8 \times 2 \times 2 \times 5$

$F = 160$ N

Q.27 Two persons pull a wire towards themselves. Each person exerts a force of 200 N on the wire. Young's modulus of the material of wire is 1×10^{11} N m^{-2}. Original length of the wire is 2 m and the area of cross section is 2 cm^2. The wire will extend in length by _____ μm.

Ans. **[20]**

Sol.



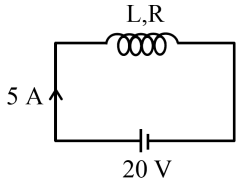
$$\frac{F}{A} = Y\frac{\Delta\ell}{\ell} \Rightarrow \Delta\ell = \frac{F\ell}{AY}$$

$$\Delta\ell = \frac{200 \times 2}{2 \times 10^{-4} \times 10^{11}} = 2 \times 10^{-5} = 20\mu m$$

Q.28 When a coil is connected across a 20 V dc supply, it draws a current of 5 A. When it is connected across 20 V, 50 Hz ac supply, it draws a current of 4 A. The self inductance of the coil is _____ mH.
(Take $\pi = 3$)

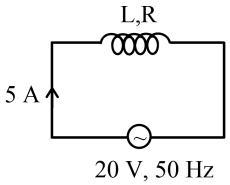
Ans. **[10]**

Sol. **Case-I :**



$$i = \frac{20}{R} \Rightarrow R = 4\Omega$$

Case-II :



$$i = \frac{20}{Z}$$

$$4 = \frac{20}{\sqrt{R^2 + X_L^2}} \Rightarrow \sqrt{R^2 + X_L^2} = 5$$

$$R^2 + X_L^2 = 25$$

$\Rightarrow X_L = 3\Omega$

$$L = \frac{3}{2\pi f} = \frac{1}{2 \times 50} = \frac{1000}{100} \text{ mH}$$

$L = 10$ mH

Q.29 The position, velocity and acceleration of a particle executing simple harmonic motion are found to have magnitudes of 4 m, 2 ms^{-1} and 16 ms^{-2} at a certain instant. The amplitude of the motion is $\sqrt{x}$ m where x is _____.

Ans. **[17]**

Sol. $x = 4m$, $V = 2$ m/s, $a = 16$ m/s^2

$|a| = \omega^2 x$

$\Rightarrow 16 = \omega^2(4)$

$\omega = 2$ rad/s

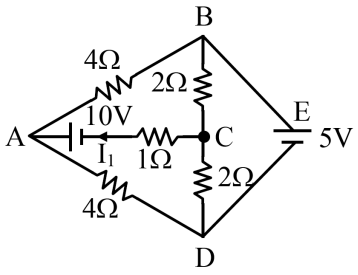
$v = \omega\sqrt{A^2 - x^2}$

$A = \sqrt{\frac{v^2}{\omega^2} + x^2}$

$\Rightarrow A = \sqrt{\frac{4}{4} + 16}$

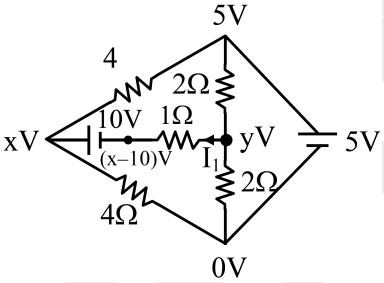
$A = \sqrt{17}$ m

Q.30 The current flowing through the 1 Ω resistor is n/10 A. The value of n is .....



Ans. **[25]**

Sol.



$$\frac{y-5}{2} + \frac{y-0}{2} + \frac{y-x+10}{1} = 0$$

$y - 5 + y + 2y - 2x + 20 = 0$

$4y - 2x + 15 = 0$ ...(i)

$$\frac{x-5}{4} + \frac{x-0}{4} + \frac{x-10-y}{1} = 0$$

$x - 5 + x + 4x - 40 - 4y = 0$

$6x - 4y - 45 = 0$ ...(i)

$-2x + 4y + 15 = 0$ ...(ii)

$4x - 30 = 0$

$x = \frac{15}{2}$ & $4y - 15 + 15 = 0$

$y = 0$

$i = \frac{y - x + 10}{1}$

$i = \frac{0 - 7.5 + 10}{1}$

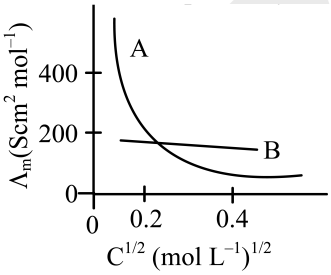
$i = 2.5A = \frac{n}{10} A$

$n = 25$

CHEMISTRY

Section-A: This section contains 20 multiple choice questions. Each question has 4 choices(1), (2), (3) and (4), out of which **ONLY ONE** is correct..

Q.31 The molar conductivity for electrolytes A and B are plotted against $C^{1/2}$ as shown below. Electrolytes A and B respectively are :



(1) A → Weak electrolyte; B → weak electrolyte
(2) A → Strong electrolyte; B → strong electrolyte
(3) A → Weak electrolyte; B → strong electrolyte
(4) A → strong electrolyte; B → weak electrolyte

Ans. **[3]**

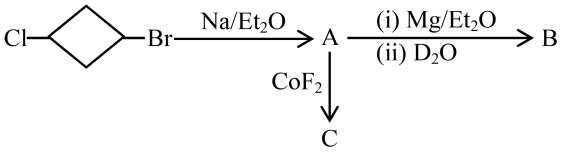
Sol. A → Weak electrolyte; B → strong electrolyte

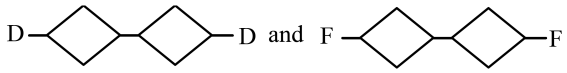
Q.32 Methods used for purification of organic compounds are based on :
(1) neither on nature of compound nor on the impurity present.
(2) nature of compound only.
(3) nature of compound and presence of impurity.
(4) presence of impurity only.

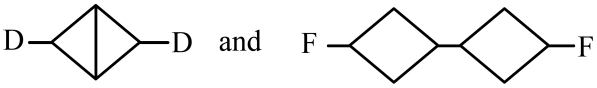
Ans. **[3]**

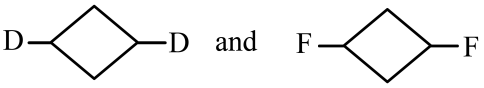
Sol. Organic compounds are purified based on their nature and impruity present in it.

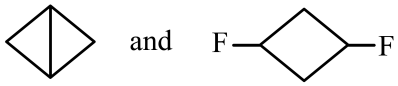
Q.33 In the following sequence of reaction, the major products B and C respectively are :



(1)  and

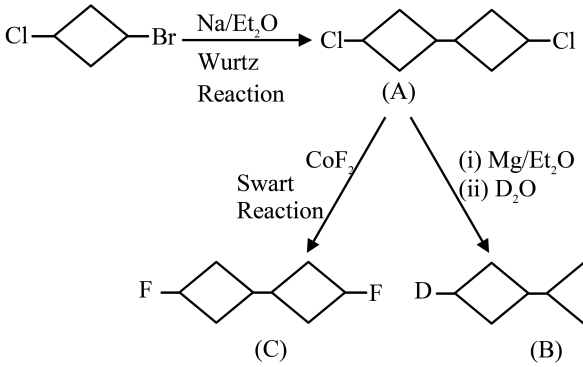
(2) 

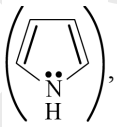
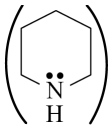
(3) 

(4) 

Ans. **[1]**

Sol.



Q.34 Correct order of basic strength of Pyrrole  , Pyridine  , and Piperidine  is

(1) Piperidine > Pyridine > Pyrrole (2) Pyrrole > Pyridine > Piperidine

(3) Pyridine > Piperidine > Pyrrole (4) Pyrrole > Piperidine >Pyridine

Ans. **[1]**

Sol. Order of basic strength is N(sp^3, localized lone pair) > N(sp^2, localized lone pair) > N(sp^2, delocalized lone pair, aromatic)

$\therefore$ Piperidine > Pyridine > Pyrrole

Q.35 In which one of the following pairs the central atoms exhibit sp^2 hybridization ?

(1) BF_3 and NO_2^- (2) NH_2^- and H_2O (3) H_2O and NO_2 (4) NH_2^- and BF_3

Ans. **[1]**

Sol. $BF_3 \rightarrow sp^2$

$NO_2^- \rightarrow sp^2$

$H_2O \rightarrow sp^3$

$NO_2 \rightarrow sp^2$

$NH_2^- \rightarrow sp^3$

Q.36 The F^- ions make the enamel on teeth much harder by converting hydroxyapatite (the enamel on the surface of teeth) into much harder fluoroapatite having the formula.

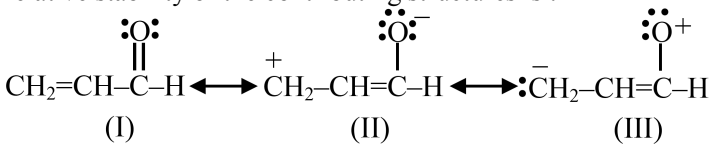
(1) $[3(Ca_3(PO_4)_2).CaF_2]$ (2) $[3(Ca_2(PO_4)_2).Ca(OH)_2]$

(3) $[3(Ca_3(PO_4)_3).CaF_2]$ (4) $[3(Ca_3(PO_4)_2).Ca(OH)_2]$

Ans. **[1]**

Sol. Fluoroapatite $\Rightarrow [3Ca_3(PO_4)_2.CaF_2]$

Q.37 Relative stability of the contributing structures is :



(I) (II) (III)

(1) (I) > (III) > (II) (2) (I) > (II) > (III) (3) (II) > (I) > (III) (4) (III) > (II) > (I)

Ans. **[2]**

Sol. (1) Neutral structures are more stable than charged ones. Therefore I is more stable than II and III.

(2) +ve charge on less electronegative atom is more stable i.e., $C^{\oplus}$ is more stable than $O^{\oplus}$

$\therefore$ Order is I > II > III

Q.38 Given below are two statements :

Statement (I) : The oxidation state of an element in a particular compound is the charge acquired by its atom on the basis of electron gain enthalpy consideration from other atoms in the molecule.

Statement (II) : $p\pi$-$p\pi$ bond formation is more prevalent in second period elements over other periods.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Both Statement I and Statement II are correct

(4) Statement I is incorrect but Statement II is correct

Ans. **[4]**

Sol. Oxidation state of an element in a particular compound is defined by the charge acquired by its atom on the basis of electronegativity consideration from other atoms in molecule.

Q.39 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)** :

Assertion (A) : SN_2 reaction of $C_6H_5CH_2Br$ occurs more readily than the SN_2 reaction of CH_3CH_2Br.

Reason (R) : The partially bonded unhybridized p-orbital that develops in the trigonal bipyramidal transition state is stabilized by conjugation with the phenyl ring.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) (A) is not correct but (R) is correct

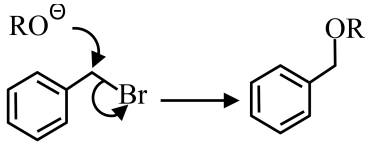
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(4) (A) is correct but (R) is not correct

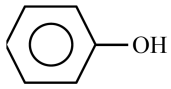
Ans. **[3]**

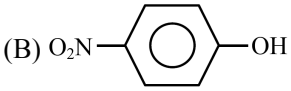
Sol. The benzyl group acts in much the same way using the π-system of the benzene ring for conjugation with the p-orbital in the transition state.

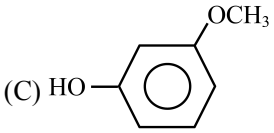


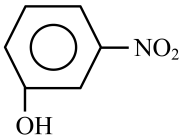
benzyl bromide

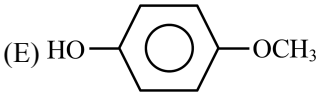
Q.40 For the given compounds, the correct order of increasing pK_a value :

(A) 

(B) 

(C) 

(D) 

(E) 

(1) (E) < (D) < (C) < (B) < (A)

(2) (D) < (E) < (C) < (B) < (A)

(3) (E) < (D) < (B) < (A) < (C)

(4) (B) < (D) < (C) < (A) < (E)

Ans. **[4]**

Sol. Acidic strength order :-

B > D > C > A > E

Correct pKa Order :

B < D < C < A < E

Q.41 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Both rhombic and monoclinic sulphur exist as S_8 while oxygen exists as O_2.

Reason (R) : Oxygen forms pπ-pπ multiple bonds with itself and other elements having small size and high electronegativity like C, N, which is not possible for sulphur.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct but (R) is correct.

Ans. **[3]**

Sol. Oxygen can form 2pπ-2pπ multiple bond with itself due to its small size while sulphur cannot form multiple bond with itself as 3pπ-3pπ bond will be unstable due to large size of sulphur, but sulphur can form multiple bond with small size atom like C and N.

eg. S = C = S

$S = C = N^- \leftrightarrow S^{\oplus}C \equiv N$

Q.42 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A): The total number of geometrical isomers shown by $[Co(en)_2Cl_2]^+$ complex ion is three

Reason (R) : $[Co(en)_2Cl_2]^+$ complex ion has an octahedral geometry.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

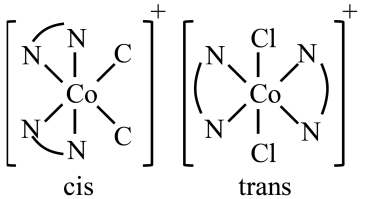
(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

Ans. **[3]**

Sol. $[Co(en)_2Cl_2]^+$ has octahedral geometry with two geometrical isomers.



cis trans

Q.43 The electronic configuration of Cu(II) is $3d^9$ whereas that of Cu(I) is $3d^{10}$. Which of the following is correct ?

(1) Cu(II) is less stable

(2) Stability of Cu(I) and Cu(II) depends on nature of copper salts

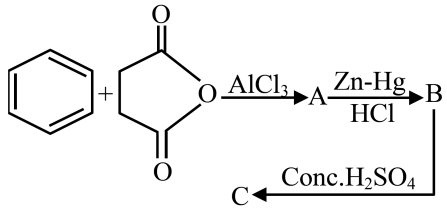
(3) Cu(II) is more stable

(4) Cu(I) and Cu(II) are equally stable

Ans. **[3]**

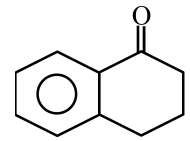
Sol. Cu(II) is more stable than Cu(I) because hydration energy of Cu^{+2} ion compensate IE_2 of Cu.

Q.44

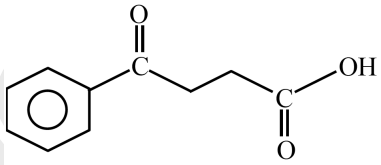


What is the structure of C ?

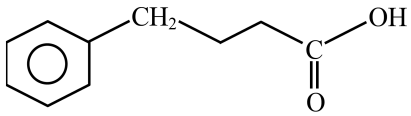
(1)



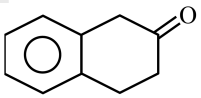
(2)



(3)

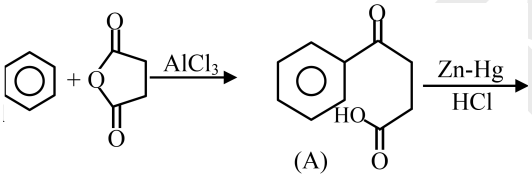


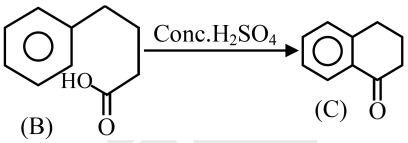
(4)



Ans. **[1]**

Sol.





Q.45 Compare the energies of following sets of quantum numbers for multielectron system.

(A) $n = 4, \ell = 1$ (B) $n = 4, \ell = 2$ (C) $n = 3, \ell = 1$ (D) $n = 3, \ell = 2$

(E) $n = 4, \ell = 0$

Choose the correct answer from the options given below :

(1) (B) > (A) > (C) > (E) > (D) (2) (E) > (C) < (D) < (A) < (B)

(3) (E) > (C) > (A) > (D) > (B) (4) (C) < (E) < (D) < (A) < (B)

Ans. **[4]**

Sol. Energy level can be determined by comparing $(n + \ell)$ values

(A) $n = 4, \ell = 1 \Rightarrow (n + \ell) = 5$

(B) $n = 4, \ell = 2 \Rightarrow (n + \ell) = 6$

(C) $n = 3, \ell = 1 \Rightarrow (n + \ell) = 4$

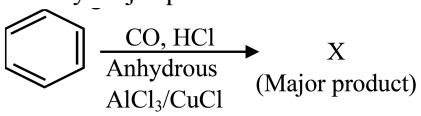
(D) $n = 3, \ell = 2 \Rightarrow (n + \ell) = 5$

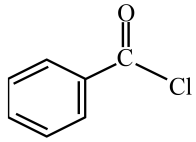
(E) $n = 4, \ell = 0 \Rightarrow (n + \ell) = 4$

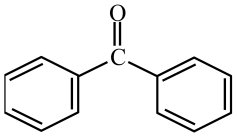
For same value of $(n + \ell)$, orbital having higher value of n, will have more energy.

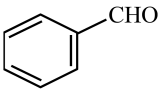
(B) > (A) > (D) > (E) > (C)

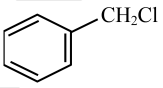
Q.46 Identify major product "X" formed in the following reaction :



(1) 

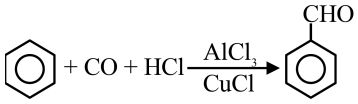
(2) 

(3) 

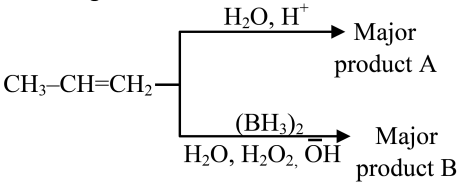
(4) 

Ans. **[3]**

Sol. This is Gattermann-Koch reaction



Q.47 Identify the product A and product B in the following set of reactions.



(1) A-$CH_3CH_2CH_2$–OH, B-$CH_3CH_2CH_2$–OH

(2) A-$CH_3CH_2CH_2$–OH, B-$CH_3\underset{\underset{OH}{|}}{CH}–CH_3$

(3) A- $CH_3 – \underset{\underset{OH}{|}}{CH} – CH_3$, B-$CH_3CH_2CH_2$–OH

(4) A-$CH_3CH_2CH_3$, B-$CH_3CH_2CH_3$

Ans. **[3]**

Sol. **(1) Hydration Reaction :**

$$CH_3 - CH = CH_2 + H^+ \longrightarrow \underset{\text{(More stable)}}{CH_3 - \overset{+}{C}H - CH_3}$$

$$CH_3–\overset{+}{C}H–CH_3 + H_2O \longrightarrow \underset{\underset{OH}{|}}{(CH_3–CH–CH_3)} + H^+$$

(A)

(2) Hydroboration Oxidation Reaction :

$$3CH_3–CH=CH_2 + B_2H_6 \xrightarrow{THF} 2(CH_3CH_2CH_2)_3B$$

$$(CH_3CH_2\,CH_2)_3\,B + 3H_2O_2 \xrightarrow{OH^-} \underset{\text{(B)}}{3CH_3CH_2\,CH_2OH} + H_3BO_3$$

Q.48 On reaction of Lead Sulphide with dilute nitric acid which of the following is not formed ?

(1) Lead nitrate (2) Sulphur (3) Nitric oxide (4) Nitrous oxide

Ans. **[4]**

Sol. $PbS + HNO_3 \rightarrow Pb(NO_3)_2 + NO + S + H_2O$

Nitrous oxide (N_2O) is not formed during the reaction.

Q.49 Identify the **incorrect** statements regarding primary standard of titrimetric analysis

(A) It should be purely available in dry form.

(B) It should not undergo chemical change in air.

(C) It should be hygroscopic and should react with another chemical instantaneously and stoichiometrically.

(D) It should be readily soluble in water.

(E) $KMnO_4$ & NaOH can be used as primary standard.

Choose the **correct** answer from the options given below :

(1) (C) and (D) only (2) (B) and (E) only (3) (A) and (B) only (4) (C) and (E) only

Ans. **[4]**

Sol. $KMnO_4$ & NaOH $\rightarrow$ Secondary standard.

Primary standard should not be Hygroscopic.

Q.50 0.05M $CuSO_4$ when treated with 0.01M $K_2Cr_2O_7$ gives green colour solution of $Cu_2Cr_2O_7$. The [SPM : Semi Permeable Membrane]

$K_2Cr_2O_7$	$CuSO_4$

Side X SPM Side Y

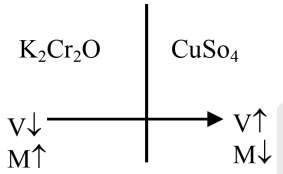
Due to osmosis :

(1) Green colour formation observed on side Y. (2) Green colour formation observed on side X.

(3) Molarity of $K_2Cr_2O_7$ solution is lowered. (4) Molarity of $CuSO_4$ solution is lowered.

Ans. **[4]**

Sol. Only solvent Molecules are allowed to pass through the SPM.



Section-B: Numerical Value Type Questions: This section contains 10 Numerical based questions. Attempt any 5 questions out of 10. The answer to each question should be rounded-off to the nearest integer.

Q.51 The heat of solution of anhydrous $CuSO_4$ and $CuSO_4 \cdot 5H_2O$ are –70 kJ mol^{-1} and +12 kJ mol^{-1} respectively. The heat of hydration of $CuSO_4$ to $CuSO_4 \cdot 5H_2O$ is –x kJ. The value of x is_____.

Ans. **[82]**

Sol.

$$(1)\quad CuSO_4(s) + 5H_2O \xrightarrow{x} CuSO_4.5H_2O$$

$$(2)\quad CuSO_4.5H_2O + H_2O \xrightarrow{12} CuSO_4(aq)$$

$$CuSO_4 + H_2O \xrightarrow{-70} CuSO_4(aq)$$

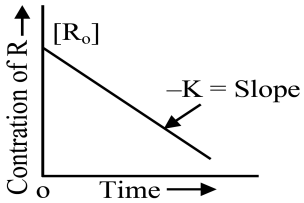
from (1) & (2)

$–70 = x + 12 \Rightarrow \quad x = –82$

Q.52 Given below are two statements :

Statement I : The rate law for the reaction A + B $\rightarrow$ C is rate (r) = $k[A]^2[B]$. When the concentration of both A and B is doubled, the reaction rate is increased "x" times.

Statement II :



The figure is showing "the variation in concentration against time plot" for a "y" order reaction.

The value of x + y is _____.

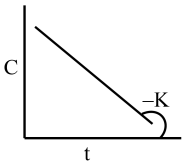
Ans. **[8]**

Sol. $r = K[A]^2|B|$

if conc. are doubled

$r' = K[2A]^2[2B]^1$

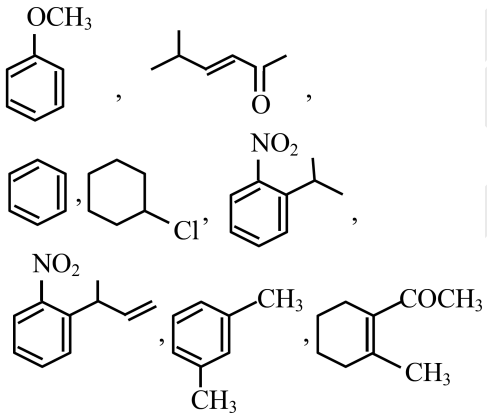
$r' = 8r \Rightarrow x = 8$



$\Rightarrow$ Zero order, y = 0

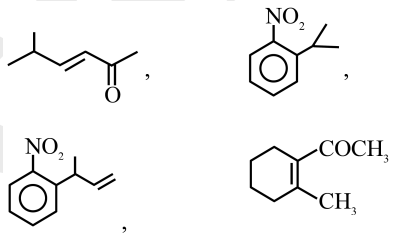
x + y = 8

Q.53 How many compounds among the following compounds show inductive, mesomeric as well as hyperconjugation effects?



Ans. **[4]**

Sol.



Q.54 The standard reduction potentials at 298 K for the following half cells are given below :

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$, $E° = 1.33V$

$Fe^{3+}(aq) + 3e^- \rightarrow Fe$, $E° = -0.04V$

$Ni^{2+}(aq) + 2e^- \rightarrow Ni$, $E° = -0.25V$
$Ag^+(aq) + e^- \rightarrow Ag$, $E° = 0.80V$
$Au^{3+}(aq) + 3e^- \rightarrow Au$, $E° = 1.40V$
Consider the given electrochemical reactions,
The number of metal(s) which will be oxidized be $Cr_2O_7^{2-}$, in aqueous solution is _____.

Ans. **[3]**

Sol. Fe, Ni, Ag will be oxidized due to lower S.R.P.

Q.55 When equal volume of 1M HCl and 1M H_2SO_4 are separately neutralised by excess volume of 1M NaOH solution. x and y kJ of heat is liberated respectively. The value of y/x is _____.

Ans. **[2]**

Sol. $H^+ + OH^- \rightarrow H_2O \Rightarrow x$
$2H^+ + 2OH^- \rightarrow 2H_2O \Rightarrow 2x = y \Rightarrow y/x = 2$

Q.56 Molarity (M) of an aqueous solution containing x g of anhyd. $CuSO_4$ in 500 mL solution at 32 °C is 2×10^{-1} M. Its molality will be _____ $\times 10^{-3}$ m. (nearest integer).
[Given density of the solution = 1.25 g/mL.]

Ans. **[164]**

Sol. $M_{solution} = v_{solution} \times d_{solution}$
$= 500 \times 1.25 = 625g$
Mass of solute (x) $= 0.2 \times 0.5 \times 159.5 = 15.95$
$n_{solute} = 0.1$,
Mass of solvent = Mass of solution – Mass of solute
$= 625 - 15.95 = 609.05$

$$m = \frac{0.1}{\frac{609.05}{1000}} \Rightarrow m = 0.164 = 164 \times 10^{-3}$$

Q.57 The total number of species from the following in which one unpaired electron is present, is _____.
$N_2, O_2, C_2^-, O_2^-, O_2^{2-}, H_2^+, CN^-, He_2^+$

Ans. **[4]**

Sol. One unpaired e^- is present in :
$C_2^-; O_2^-; H_2^+; He_2^+$

Q.58 Number of ambidentate ligands among the following is _____.
$NO_2^-, SCN^-, C_2O_4^{2-}, NH_3, CN^-, SO_4^{2-}, H_2O$

Ans. **[3]**

Sol. Ligands which have two different donor sites but at a time connects with only one donor site to central metal are ambidentate ligands.
Ambidentate ligands are NO_2^-, SCN^-, CN^-

Q.59 Total number of essential amino acid among the given list of amino acids is _____.
Arginine, Phenylalanine, Aspartic acid, Cysteine, Histidine, Valine, Proline

Ans. **[4]**

Sol. Essential Amino acids are :-
Arginine, Phenylalanine, Histidine, Valine

Q.60 Number of colourless lanthanoid ions among the following is _____.
$Eu^{3+}, Lu^{3+}, Nd^{3+}, La^{3+}, Sm^{3+}$

Ans. **[2]**

Sol. La^{+3} – $[Xe]4f^0$
Nd^{+3} – $[Xe]4f^3$
Sm^{+3} – $[Xe]4f^5$
Eu^{+3} – $[Xe]4f^6$
Lu^{+3} – $[Xe]4f^{14}$
La^{+3} and Lu^{+3} do not show any colour because no unpaired electron is present.

MATHEMATICS

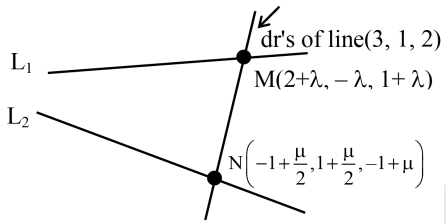
Section-A: This section contains 20 multiple choice questions. Each question has 4 choices(1), (2), (3) and (4), out of which **ONLY ONE** is correct..

Q.61 Let the line L interesct the lines $x - 2 = -y = z - 1$, $2(x + 1) = 2(y - 1) = z + 1$ and be parallel to the line $\frac{x-2}{3} = \frac{y-1}{1} = \frac{z-2}{2}$. Then which of the following points lies on L?

(1) $\left(-\frac{1}{3}, 1, 1\right)$ (2) $\left(-\frac{1}{3}, 1, -1\right)$ (3) $\left(-\frac{1}{3}, -1, -1\right)$ (4) $\left(-\frac{1}{3}, -1, 1\right)$

Ans. **[2]**

Sol.



$L_1 : \frac{x-2}{1} = \frac{y}{-1} = \frac{z-1}{1} = \lambda$

$L_2 : \frac{x+1}{1/2} = \frac{y-1}{1/2} = \frac{z+1}{1} = \mu$

dr of line MN will be

$< 3 + \lambda - \frac{\mu}{2}, -1 - \lambda - \frac{\mu}{2}, 2 + \lambda - \mu >$ & it will be proportional to $< 3, 1, 2 >$

$\therefore \frac{3+\lambda-\frac{\mu}{2}}{3} = \frac{-1-\lambda-\frac{\mu}{2}}{1} = \frac{2+\lambda-\mu}{2}$

$4\lambda + \mu = -6$ $\quad$ $4 + 3\lambda = 0$

$\Rightarrow \lambda = -\frac{4}{3}$ & $\mu = -\frac{2}{3}$

$\therefore$ Coordinate of M will be $< \left(\frac{2}{3}, \frac{4}{3}, -\frac{1}{3}\right)$ and equation of required line will be

$\frac{x-\frac{2}{3}}{3} = \frac{y-\frac{4}{3}}{1} = \frac{z+\frac{1}{3}}{2} = k$

So any point on this line will be

$\left(\frac{2}{3}+3k, \frac{4}{3}+k, -\frac{1}{3}+2k\right)$

$\because \frac{2}{3}+3k=-\frac{1}{3} \Rightarrow k=-\frac{1}{3}$

$\therefore$ Point lie on the line for

$k=-\frac{1}{3}$ is $\left(-\frac{1}{3}, 1, -1\right)$

Q.62 The parabola $y^2 = 4x$ divides the area of the circle $x^2 + y^2 = 5$ in two parts. The area of the smaller part is equal to

(1) $\frac{2}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ (2) $\frac{1}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

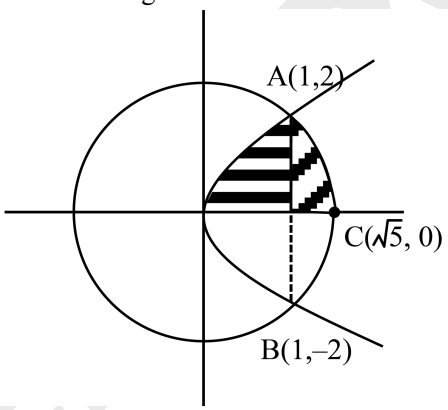
(3) $\frac{1}{3}+\sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ (4) $\frac{2}{3}+\sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

Ans. **[1]**

Sol. $y^2 = 4x$

$x^2 + y^2 = 5$

$\therefore$ Area of shaded region as shown in the figre will be



$A_1 = \int_0^1 \sqrt{4x}\,dx + \int_1^{\sqrt{5}} \sqrt{5-x^2}\,dx$

$= \frac{4}{3}.\left[x^{3/2}\right]_0^1 + \left[\frac{x}{2}\sqrt{5-x^2}+\frac{5}{2}\sin^{-1}\frac{x}{\sqrt{5}}\right]_1^{\sqrt{5}}$

$= \frac{1}{3}+\frac{5\pi}{4}-\frac{5}{2}\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$

$\therefore$ Required area = $2A_1$

$= \frac{2}{3}+\frac{5\pi}{2}-5\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$

$= \frac{2}{3}+5\left(\frac{\pi}{2}-\sin^{-1}\frac{1}{\sqrt{5}}\right)$

$= \frac{2}{3}+5\cos^{-1}\frac{1}{\sqrt{5}}$

$= \frac{2}{3} + 5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

Q.63 The solution curve, of the differential equation $2y\frac{dy}{dx} + 3 = 5\frac{dy}{dx}$, passing through the point (0, 1) is a conic, whose vertex lies on the line :

(1) $2x + 3y = 9$ (2) $2x + 3y = -9$ (3) $2x + 3y = -6$ (4) $2x + 3y = 6$

Ans. **[1]**

Sol. $(2y - 5)\frac{dy}{dx} = -3$

$(2y - 5)dy = -3\,dx$

$2.\ \frac{y^2}{2} - 5y = -3x + \lambda$

$\because$ Curve passes through (0, 1)

$\Rightarrow \lambda = -4$

$\because$ Curve will be

$\left(y - \frac{5}{2}\right)^2 = 3\left(x - \frac{3}{4}\right)$

$\therefore$ Vertex of parabola will be $\left(\frac{3}{4}, \frac{5}{2}\right)$

$\because\ 2x + 3y = 9$

Q.64 A ray of light coming form the point P(1, 2) gets reflected from the point Q on the x-axis and then passes through the point R(4, 3). If the point S(h, k) is such that PQRS is a parallelogram, then hk^2 is equal to

(1) 80 (2) 90 (3) 60 (4) 70

Ans. **[4]**

Sol.

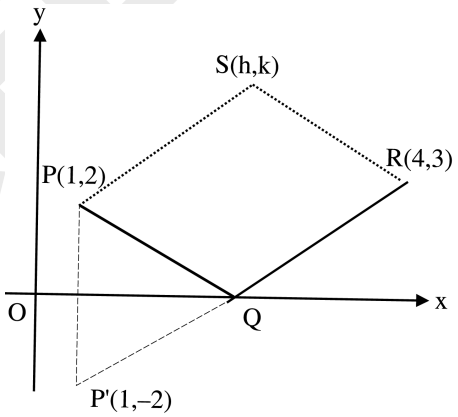


Image of P wrt x-axis will be P′ (1, –2) equation of line joining P′R will be

$y - 3 = \frac{5}{3}(x - 4)$

Above line will meet x-axis at Q where

$y = 0 \Rightarrow x = \frac{11}{5}$

$\therefore Q\left(\frac{11}{5}, 0\right)$

$\because$ PQRS is parallelogram so their diagonals will bisects each other

$\Rightarrow \frac{4+1}{2} = \frac{\frac{11}{5}+h}{2}$ & $\frac{2+3}{2} = \frac{k+0}{2}$

$\Rightarrow h = \frac{14}{5}$ & $k = 5$

$\therefore hk^2 = \frac{14}{5} \times 5^2 = 70$

Q.65 Let $\lambda, \mu \in R$. If the system of equations

$3x + 5y + \lambda z = 3$

$7x + 11y - 9z = 2$

$97x + 155y - 189z = \mu$

has infinitely many solutions, then $\mu + 2\lambda$ is equal to

(1) 25 (2) 24 (3) 27 (4) 22

Ans. **[1]**

Sol. $3x + 5y + \lambda z = 3$

$7x + 11y - 9z = 2$

$97x + 155y - 189z = \mu$

$93x + 155y + 31\lambda z = 93$

$97x + 155y - 189z = \mu$

$- \quad - \quad + \quad -$

$-4x + (31\lambda + 189)z = 93 - \mu$

$1085x + 1705y - 1395z = 310$

$1067x + 1705y - 2079z = 11\mu$

$- \quad - \quad + \quad -$

$18x + 684z = 310 - 11\mu$

$-36x + 9(31\lambda + 189)z = 9(93 - \mu)$

$36x + 1368z = 2(310 - 11\mu)$

$(279\lambda + 3069)z = 1457 - 31\mu$

for infinite solutions

$\lambda = \frac{-3069}{279} = \frac{-341}{31}$

$\mu = \frac{1457}{31}$

$\mu + 2\lambda = \frac{1457 - 682}{31} = \frac{775}{31} = 25$

Q.66 The coefficient of x^{70} in $x^2(1+x)^{98} + x^3(1+x)^{97} + x^4(1+x)^{96} + ... + x^{54}(1+x)^{46}$ is ${}^{99}C_p - {}^{46}C_q$. Then a possible value to $p + q$ is

(1) 55 (2) 61 (3) 68 (4) 83

Ans. **[4]**

Sol. $x^2(1+x)^{98} + x^3(1+x^{97}) + x^4(1+x)^{96} + ... x^{54}(1+x)^{46}$

Coeff. of x^{70} : ${}^{98}C_{68} + {}^{97}C_{67} + {}^{96}C_{66} + \ldots\ldots {}^{47}C_{17} + {}^{46}C_{16}$

$= {}^{46}C_{30} + {}^{47}C_{30} + \ldots\ldots {}^{98}C_{30}$

$= ({}^{46}C_{31} + {}^{46}C_{30}) + {}^{47}C_{30} + \ldots\ldots {}^{98}C_{30} - {}^{46}C_{31}$

$= {}^{47}C_{31} + {}^{47}C_{30} + \ldots\ldots\ldots {}^{98}C_{30} - {}^{46}C_{31} \ldots\ldots\ldots$

$= {}^{99}C_{31} - {}^{46}C_{31} = {}^{99}C_{p} - {}^{46}C_{q}$

Possible values of $(p + q)$ are 62, 83, 99, 46

$\Rightarrow p + q = 83$

Q.67 Let $\int \frac{2 - \tan x}{3 + \tan x}\, dx = \frac{1}{2}(\alpha x + \log_e |\beta \sin x + \gamma \cos x|) + C$, where C is the constant of integration. Then $\alpha + \frac{\gamma}{\beta}$ is equal to

(1) 3 (2) 1 (3) 4 (4) 7

Ans. **[3]**

Sol. $\int \frac{2 - \tan x}{3 + \tan x}\, dx = \int \frac{2\cos x - \sin x}{3\cos x + \sin x}\, dx$

$2\cos x - \sin x = A(3\cos x + \sin x) + B(\cos x - 3\sin x)$

$3A + B = 2$

$A - 3B = -1$

$\Rightarrow A = \frac{1}{2}, B = \frac{1}{2}$

$\therefore \int \frac{2\cos x - \sin x}{3\cos x + \sin x}\, dx = \frac{x}{2} + \frac{1}{2}\ln|3\cos x + \sin x| + C$

$= \frac{1}{2}(x + \ln|3\cos x + \sin x|) + C$

$= \frac{1}{2}(\alpha x + \ln|\beta \sin x + \gamma \cos x|) + C$

$\alpha = 1, \beta = 1, \gamma = 3$

$\therefore \alpha + \frac{\gamma}{\beta} = 1 + \frac{3}{1} = 4$

Q.68 A variable line L passes through the point (3, 5) and intersects the positive coordinate axes at the points A and B. The minimum area of the triangle OAB, where O is the origin, is :

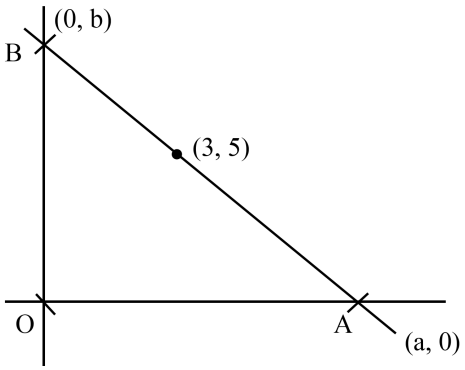
(1) 30 (2) 25 (3) 40 (4) 35

Ans. **[1]**

Sol. $\frac{x}{a} + \frac{y}{b} = 1$

$\frac{3}{a} + \frac{5}{b} = 1 \Rightarrow b = \frac{5a}{a - 3}, a > 3$

CAREER POINT



$$A = \frac{1}{2}ab = \frac{1}{2}a\frac{5a}{(a-3)} = \frac{5}{2}\cdot\frac{a^2}{a-3}$$

$$= \frac{5}{2}\left(\frac{a^2-9+9}{a-3}\right)$$

$$= \frac{5}{2}\left(a+3+\frac{9}{a-3}\right)$$

$$= \frac{5}{2}\left(a+3+\frac{9}{a-3}+6\right) \geq 30$$

Q.69 Let $|\cos\theta \cos(60-\theta) \cos(60+\theta)| \leq \frac{1}{8}$, $\theta \in [0, 2\pi]$

Then, the sum of all $\theta \in [0, 2\pi]$, where $\cos 3\theta$ attains its maximum value, is

(1) 9π (2) 18π (3) 6π (4) 15π

Ans. **[3]**

Sol. What know that

$(\cos\theta)(\cos(60^\circ-\theta)(\cos(60^\circ+\theta) = \frac{1}{4}\cos 3\theta$

So equation reduces to $\left|\frac{1}{4}\cos 3\theta\right| \leq \frac{1}{8}$

$\Rightarrow |\cos 3\theta| \leq \frac{1}{2}$

$\Rightarrow -\frac{1}{2} \leq \cos 3\theta \leq \frac{1}{2}$

$\Rightarrow$ maximum value of $\cos 3\theta = \frac{1}{2}$, here

$\Rightarrow 3\theta = 2n\pi \pm \frac{\pi}{3}$

$\theta = \frac{2n\pi}{3} \pm \frac{\pi}{9}$

As $\theta \in [0, 2\pi]$ possible values are

$\theta = \left\{\frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{11\pi}{9}, \frac{13\pi}{9}, \frac{17\pi}{9}\right\}$

Whose sum is

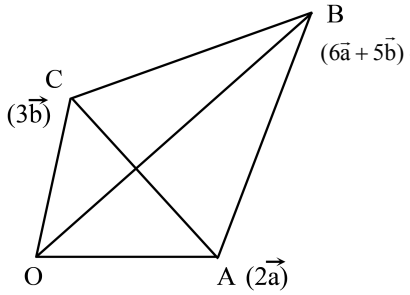
$\frac{\pi}{9}+\frac{5\pi}{9}+\frac{7\pi}{9}+\frac{11\pi}{9}+\frac{13\pi}{9}+\frac{17\pi}{9} = \frac{54\pi}{9} = 6\pi$

Q.70 Let $\overrightarrow{OA} = 2\vec{a}$, $\overrightarrow{OB} = 6\vec{a} + 5\vec{b}$ and $\overrightarrow{OC} = 3\vec{b}$, where O is the origin. If the area of the parallelogram with adjacent sides $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is 15 sq. units, then the area (in sq. unit) of the quadrilateral OABC is equal to

(1) 38 (2) 40 (3) 32 (4) 35

Ans. **[4]**

Sol.



Area of parallelogram having sides

$\overrightarrow{OA}$ & $\overrightarrow{OC}$ $= |\overrightarrow{OA} \times \overrightarrow{OC}| = |2\vec{a} \times 3\vec{b}| = 15$

$6|\vec{a} \times \vec{b}| = 15$

$\Rightarrow |\vec{a} \times \vec{b}| = \frac{5}{2}$ ...(1)

Area of quadrilateral

$OABC = \frac{1}{2}\left|\vec{d}_1 \times \vec{d}_2\right|$

$= \frac{1}{2}\left|\overrightarrow{AC} \times \overrightarrow{OB}\right| = \frac{1}{2}\left|(3\vec{b} - 2\vec{a}) \times (6\vec{a} + 5\vec{b})\right|$

$= \frac{1}{2}\left|18\vec{b} \times \vec{a} - 10\vec{a} \times \vec{b})\right| = \frac{1}{2}\left|\vec{a} \times \vec{b}\right| = 14 \times \frac{5}{2} = 35$

Q.71 If the domain of the function $f(x) = \sin^{-1}\left(\frac{x-1}{2x+3}\right)$ is $R - (\alpha, \beta)$ then $12\alpha\beta$ is equal to

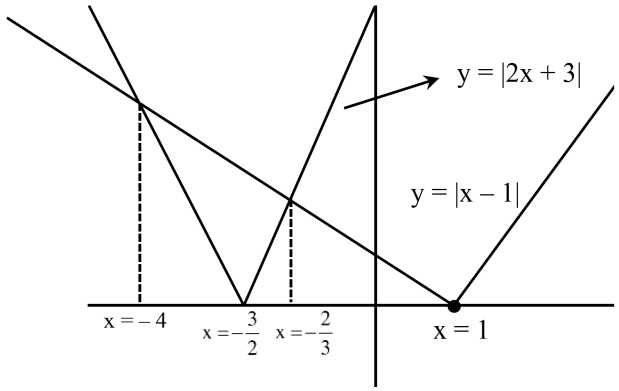
(1) 36 (2) 24 (3) 40 (4) 32

Ans. **[4]**

Sol. Domain of $f(x) = \sin^{-1}\left(\frac{x-1}{2x+3}\right)$ is

$2x + 3 \neq 0$ & $x \neq \frac{-3}{2}$ and $\left|\frac{(x-1)}{2x+3}\right| \leq 1$

$|x - 1| \leq |2x + 3|$



For $|2x + 3| \geq |x - 1|$

$x \in (-\infty, -4] \cup \left(-\frac{2}{3}, \infty\right)$

$\alpha = -4$ & $\beta = -\frac{2}{3}$: $12\alpha\beta = 32$

Q.72 If the sum of series $\frac{1}{1\cdot(1+d)} + \frac{1}{(1+d)(1+2d)} + ... + \frac{1}{(1+9d)(1+10d)}$ is equal to 5, then 50d is equal to

(1) 20 (2) 5 (3) 15 (4) 10

Ans. **[2]**

Sol. $\frac{1}{1\cdot(1+d)} + \frac{1}{(1+d)(1+2d)} + ... + \frac{1}{(1+9d)(1+10d)} = 5$

$$\frac{1}{d}\left[\frac{(1+d)-1}{1\cdot(1+d)} + \frac{(1+2d)-(1-d)}{(1+d)(1+2d)} + ...\frac{(1+10d)-(1+9d)}{(1+9d)(1+10d)}\right] = 5$$

$$\frac{1}{d}\left[\left(1-\frac{1}{1+d}\right)+\left(\frac{1}{1+d}-\frac{1}{1+2d}\right)+....\left(\frac{1}{1+9d}-\frac{1}{1+10d}\right)\right] = 5$$

$$\frac{1}{d}\left[1-\frac{1}{(1+10d)}\right] = 5$$

$$\frac{10d}{1+10d} = 5d$$

$50d = 5$

Q.73 Let $f(x) = ax^3 + bx^2 + ex + 41$ be such that $f(1) = 40$, $f'(1) = 2$ and $f''(1) = 4$. Then $a^2 + b^2 + c^2$ is equal to

(1) 62 (2) 73 (3) 54 (4) 51

Ans. **[4]**

Sol. $f(x) = ax^3 + bx^2 + cx + 41$

$f'(x) = 3ax^2 + 2bx + cx$

$\Rightarrow f'(1) = 3a + 2b + c = 2$ ...(1)

$f''(x) = 6ax + 2b$

$\Rightarrow f''(1) = 6a + 2b = 4$

$3a + b = 2$ ... (2)

(1) – (2)

$b + c = 0$ ....(3)

$f(1) = 40$

$a + b + c + 41 = 40$

use (3)

$a + 41 = 40$

by (2)

$-3 + b = 2 \Rightarrow b = 5$ & $c = -5$

$a^2 + b^2 + c^2 = 1 + 25 + 25 = 51$

Q.74 Let a circle passing through (2, 0) have its centre at the point (h, k). Let (x_c, y_c) be the point of intersection of the lines $3x + 5y = 1$ and $(2 + c)x + 5c^2y = 1$. If $h = \lim\limits_{c\to 1} x_c$ and $k = \lim\limits_{c\to 1} y_c$, then the equation of the circle is :

(1) $25x^2 + 25y^2 - 20x + 2y - 60 = 0$ (2) $5x^2 + 5y^2 - 4x - 2y - 12 = 0$

(3) $25x^2 + 25y^2 - 2x + 2y - 60 = 0$ (4) $5x^2 + 5y^2 - 4x + 2y - 12 = 0$

Ans. **[1]**

Sol. $(2 + c)x + 5c^2\left(\frac{1-3x}{5}\right) = 1$

$x = \frac{1-c^2}{2+c-3c^2}$, $y = \frac{1-3x}{5} = \frac{c-1}{5(2+c-3c^2)}$

$h = \lim_{c\to 1}\frac{(1-c)(1+c)}{(1-c)(2+3c)} = \frac{2}{5}$

$K = \lim_{c\to 1}\frac{c-1}{-5(c-1)(3c+2)} = -\frac{1}{25}$

Centre $\left(\frac{2}{25}, -\frac{1}{25}\right)$,

$r = \sqrt{\left(2-\frac{2}{5}\right)^2 + \left(0-\frac{1}{25}\right)^2} = \sqrt{\frac{64}{25}+\frac{1}{625}}$

$r = \frac{\sqrt{161}}{25}$

$\left(x-\frac{2}{5}\right)^2 + \left(y+\frac{1}{25}\right)^2 = \frac{161}{125}$

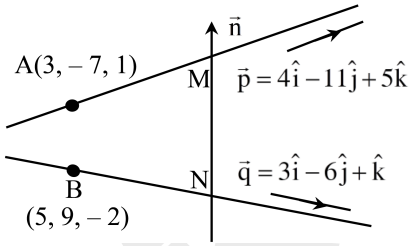
$\Rightarrow 25x^2 + 25y^2 - 20x + 2y - 60 = 0$

Q.75 The shortest distance between the line $\frac{x-3}{4} = \frac{y+7}{-11} = \frac{z-1}{5}$ and $\frac{x-5}{3} = \frac{y-9}{-6} = \frac{z+2}{1}$ is

(1) $\frac{187}{\sqrt{563}}$ (2) $\frac{178}{\sqrt{563}}$ (3) $\frac{185}{\sqrt{563}}$ (4) $\frac{179}{\sqrt{563}}$

Ans. **[1]**

Sol.



$\vec{n} = \vec{p} \times \vec{q}$

$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -11 & 5 \\ 3 & -6 & 1 \end{vmatrix} = 19\hat{i} + 11\hat{j} + 9\hat{k}$

S.d = projection of $\overrightarrow{AB}$ on $\vec{n}$

$= \left|\frac{\overrightarrow{AB}\cdot\vec{n}}{|\vec{n}|}\right| = \left|\frac{(2\hat{i}+16\hat{j}-3\hat{k})\cdot(19\hat{i}+11\hat{j}+9\hat{k})}{\sqrt{361+121+81}}\right|$

$= \frac{38+176-27}{\sqrt{563}}$

S.d. $= \frac{187}{\sqrt{563}}$

Q.76 The frequency distribution of the age of students in a class of 40 students is given below.

Age	15	16	17	18	19	*20*
No. of Student	*5*	*8*	*5*	*12*	*x*	*y*

If the mean deviation about the median is 1.25, then 4x + 5y is equal to

(1) 43 (2) 44 (3) 47 (4) 46

Ans. **[2]**

Sol. $x + y = 10$ ...(1)

Median = 18 = M

$$\text{M.D.} = \frac{\Sigma f_i \,|\, x_i - M \,|}{\Sigma f_i}$$

$$1.25 = \frac{36 + x + 2y}{40}$$

$x + 2y = 14$ ...(1)

by (1) & (2)

$x = 6, y = 4$

$\Rightarrow 4x + 5y = 24 + 20 = 44$

$\text{Age}(X_i)$	f	$\lvert x_i - M \rvert$	$f_i\lvert x_i - M \rvert$
15	5	3	*15*
16	8	2	*16*
17	5	1	*5*
18	12	0	*0*
19	x	1	*x*
20	*y*	*2*	*2y*

Q.77 The solution of the differential equation $(x^2 + y^2)dx - 5xy\,dy = 0$, y(1)=0 is :

(1) $|x^2 - 4y^2|^5 = x^2$ (2) $|x^2 - 2y^2|^6 = x$ (3) $|x^2 - 4y^2|^6 = x$ (4) $|x^2 - 2y^2|^5 = x^2$

Ans. **[1]**

Sol. $(x^2 + y^2)dx = 5xy\,dy$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 + y^2}{5xy}$$

Put $y = Vx$

$$\Rightarrow V + x\frac{dv}{dx} = \frac{1 + V^2}{5V}$$

$$\Rightarrow \frac{x\,dv}{dx} = \frac{1 - 4V^2}{5V}$$

$$\Rightarrow \int \frac{V}{1 - 4V^2}\,dV = \int \frac{dx}{5x}$$

Let $1 - 4V^2 = t$

$\Rightarrow -8V\,dV = dt$

$$\Rightarrow \int \frac{dt}{(-8)(t)} = \int \frac{dx}{5x}$$

$$\Rightarrow \frac{-1}{8}\ln|t| = \frac{1}{5}\ln|x| + \ln C$$

$\Rightarrow -5\ln|t| = 8\ln|x| + \ln K$

$\Rightarrow \ln x^8 + \ln|t^5| + \ln K = 0$

$\Rightarrow x^8|t^5| = C$

$\Rightarrow x^8|1-4V^2|^5 = C$

$\Rightarrow x^8\left|\frac{x^2-4y^2}{x^2}\right|^5 = C$

$\Rightarrow |x^2-4y^2|^5 = Cx^2$

given $y(1) = 0$

$\Rightarrow |1|^5 = C \Rightarrow C = 1$

$\Rightarrow |x^2-4y^2|^5 = x^2$

Q.78 Let three vectors $\vec{a} = \alpha\hat{i}+4\hat{j}+2\hat{k}$, $\vec{b} = 5\hat{i}+3\hat{j}+4\hat{k}$, $\vec{c} = x\hat{i}+y\hat{j}+z\hat{k}$ from a triangle such that $\vec{c} = \vec{a}-\vec{b}$ and the area of the triangle is $5\sqrt{6}$. If α is a positive real number, then $|\vec{c}|^2$ is

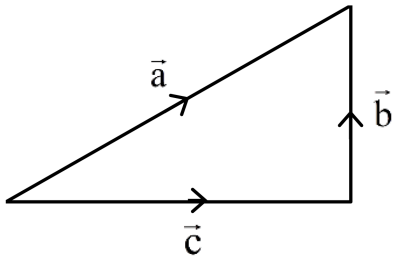
(1) 16 (2) 14 (3) 12 (4) 10

Ans. **[2]**

Sol. $\vec{c} = \vec{a}-\vec{b}$

$\Rightarrow (x, y, z) = (\alpha-5, 1, -2)$

$\Rightarrow x = \alpha-5, y = 1, z = -2$ ....(1)



Area of $\Delta = 5\sqrt{6}$ (given)

$\frac{1}{2}|\vec{a}\times\vec{c}| = 5\sqrt{6}$

$$\left|\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ \alpha & 4 & 2\\ 8 & 1 & -2\end{vmatrix}\right| = 10\sqrt{6}$$

$\Rightarrow |-10\hat{i}-\hat{j}(-2\alpha-2x)+\hat{k}(\alpha-4x)| = 10\sqrt{6}$

$\Rightarrow (2\alpha+2\alpha-10)^2+(\alpha-4\alpha+20)^2 = 500$

$\Rightarrow (4\alpha-10)^2+(20-3\alpha)^2 = 500$

$\Rightarrow 25\alpha^2-80\alpha-120\alpha = 0$

$\Rightarrow \alpha(25\alpha-200) = 0$

$\Rightarrow \alpha = 8$ (given α is +ve number)

$\Rightarrow x = \alpha-5 = 3$

$|\vec{c}|^2 = x^2+y^2+z^2$

$= 9+1+4$

$= 14$

Q.79 Let α, β be the roots of the equation $x^2 + 2\sqrt{2}x - 1 = 0$. The quadratic eqution, whose roots are $\alpha^4 + \beta^4$ and $\frac{1}{10}(\alpha^6 + \beta^6)$, is

(1) $x^2 - 190x + 9466 = 0$ (2) $x^2 - 195x + 9466 = 0$

(3) $x^2 - 195x + 9506 = 0$ (4) $x^2 - 180x + 9506 = 0$

Ans. **[3]**

Sol. $x^2 + 2\sqrt{2}x - 1 = 0$

$\alpha + \beta = -2\sqrt{2}$

$\alpha\beta = -1$

$\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$

$= ((\alpha + \beta)^2 - 2\alpha\beta)^2 - 2(\alpha\beta)^2$

$= (8 + 2)^2 - 2(-1)^2$

$= 100 - 2 = 98$

$=\alpha^6+\beta^6=(\alpha^3+\beta^3)^2-2\alpha^3\beta^3$

$= ((\alpha + \beta)((\alpha + \beta)^2 - 3\alpha\beta)^2 - 2(\alpha\beta)^3$

$= (-2\sqrt{2}(8 + 3))^2 + 2$

$= (8)(121) + 2 = 970$

$\frac{1}{10}(\alpha^6 + \beta^6) = 97$

$x^2 - (98 + 97)x + (98)(97) = 0$

$\Rightarrow x^2 - 195x + 9506 = 0$

Q.80 Let $f(x) = x^2 + 9$, $g(x) = \frac{x}{x-9}$ and $a = fog(10)$, $b = gof(3)$. If e and ℓ denote the eccentricity and the length of the latus rectum of the ellipse $\frac{x^2}{a} + \frac{y^2}{b} = 1$, then $8e^2 + \ell^2$ is equal to

(1) 16 (2) 8 (3) 6 (4) 12

Ans. **[2]**

Sol. $f(x) = x^2 + 9$ $g(x) = \frac{x}{x-9}$

$a = f(g(10)) = f\left(\frac{10}{10-9}\right)$

$= f(10) = 109$

$b = g(f(3)) = g(9 + 9)$

$= g(18) = \frac{18}{9} = 2$

$E : \frac{x^2}{109} + \frac{y^2}{2} = 1$

$e^2 = 1 - \frac{2}{109} = \frac{107}{109}$

$\ell = \frac{2(2)}{\sqrt{109}} = \frac{4}{\sqrt{109}}$

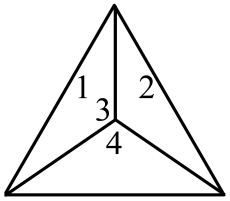
$8e^2 + \ell^2 = \frac{8(107)}{109} + \frac{16}{109} = 8$

Section-B: Numerical Value Type Questions: This section contains 10 Numerical based questions. Attempt any 5 questions out of 10. The answer to each question should be rounded-off to the nearest integer.

Q.81 Let a, b and c denote the outcome of three independent rolls of a fair tetrahedral die, whose four faces are marked 1, 2, 3, 4. If the probability that $ax^2 + bx + c = 0$ has all real roots is m/n, gcd(m, n) = 1, then m + n is equal to .....

Ans. **[19]**

Sol. $a, b, c \in \{1, 2, 3, 4\}$



Tetrahedral dice

$ax^2 + bx + c = 0$

has all real roots

$\Rightarrow D \geq 0$

$\Rightarrow b^2 - 4ac \geq 0$

Let $b = 1 \Rightarrow 1 - 4ac \geq 0$ (Not feasible)

$b = 2 \Rightarrow 4 - 4ac \geq 0$

$1 \geq ac \Rightarrow a = 1, c = 1,$

$b = 3 \Rightarrow 9 - 4ac \geq 0$

$\frac{9}{4} \geq ac$

$\Rightarrow a = 1, c = 1$

$\Rightarrow a = 1, c = 2$

$\Rightarrow a = 2, c = 1$

$b = 4 \Rightarrow 16 - ac \geq 0$

$4 \geq ac$

$\Rightarrow a = 1, c = 1$

$\Rightarrow a = 1, c = 2 \Rightarrow a = 2, c = 1$

$\Rightarrow a = 1, c = 3 \Rightarrow a = 3, c = 1$

$\Rightarrow a = 1, c = 4 \Rightarrow a = 4, c = 1$

$\Rightarrow a = 2, c = 2$

$$\text{probability} = \frac{12}{(4)(4)(4)} = \frac{3}{16} = \frac{m}{n}$$

$m + n = 19$

Q.82 The sum of the square of the modulus of the elements in the set $\{z = a + ib; a, b \in Z, z \in C, |z - 1| \leq 1, |z - 5| \leq |z - 5i|\}$ is .....

Ans. **[9]**

Sol. $|z - 1| \leq 1$

$\Rightarrow |(x - 1) + iy| \leq 1$

$\Rightarrow \sqrt{(x-1)^2 + y^2} \leq 1$

$\Rightarrow (x - 1)^2 + y^2 \leq 1$ ....(1)

Also $|z - 5| \leq |z - 5i|$

$(x-5)^2 + y^2 \le x^2 + (y-5)^2$

$-10x \le -10y$

$\Rightarrow x \ge y$ ...(2)

Solving (1) and (2)

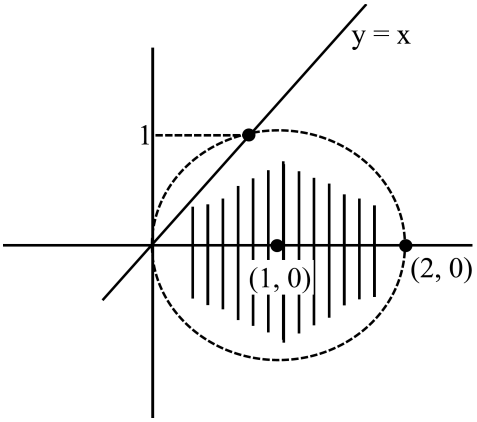
$\Rightarrow (x-1)^2 + x^2 = 1$

$\Rightarrow 2x^2 - 2x = 0$

$\Rightarrow x(x-1) = 0$

$\Rightarrow x = 0$ or $x = 1$

$y = 0$ or $y = 1$



Given x, y, $\in$ I

Points (0, 0), (1, 0), (2, 0), (1, 1), (1, –1) to find

$|z_1|^2 + |z_2|^2 + |z_3|^2 + |z_4|^2 + |z_5|^2$

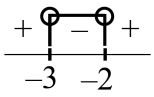
$= 0 + 1 + 4 + 1 + 1 + 1 + 1 = 9$

Q.83 Let the set of all positive values of λ, for which the point of local minimum of the function $(1 + x(\lambda^2 - x^2))$ satisfies $\frac{x^2 + x + 2}{x^2 + 5x + 6} < 0$, be (α, β). Then $\alpha^2 + \beta^2$ is equal to .....

Ans. **[39]**

Sol. $\frac{x^2 + x + 2}{x^2 + 5x + 6} < 0$

$\Rightarrow \frac{1}{(x+2)(x+3)} < 0$



$x \in (-3, -2)$ ...(1)

$f(x) = 1 + x(\lambda^2 - x^2)$

Finding local minima

$f'(x) = (\lambda^2 - x^2) + (-2x) \cdot x$

Put $f'(x) = 0$

$\Rightarrow \lambda^2 = 3x^2$

$\Rightarrow x = \pm \frac{\lambda}{\sqrt{3}}$

$$\underset{\text{Local min}}{-} \quad \underset{\text{Local max}}{+} \quad -$$
$$\frac{-\lambda}{\sqrt{3}} \qquad \frac{\lambda}{\sqrt{3}}$$

We want local min

$\Rightarrow x = \frac{-\lambda}{\sqrt{3}}$

from (1)

$x \in (-3, -2)$

$-3 < \frac{-\lambda}{\sqrt{3}} < -2$

$3\sqrt{3} > \lambda > 2\sqrt{3}$

$\alpha = 2\sqrt{3}$, $\beta = 3\sqrt{3}$

$\alpha^2 + \beta^2 = 12 + 27 = 39$

Q.84 Let $\lim\limits_{n\to\infty}\left(\frac{n}{\sqrt{n^4+1}} - \frac{2n}{(n^2+1)\sqrt{n^4+1}} + \frac{n}{\sqrt{n^4+16}} - \frac{8n}{(n^2+4)\sqrt{n^4+16}} + ... + \frac{n}{\sqrt{n^4+n^4}} - \frac{2n\cdot n^2}{(n^2+n^2)\sqrt{n^4+n^4}}\right)$ be $\frac{\pi}{k}$, using only the principal values of the inverse trigonometric functions. Then k^2 is equal to .....

Ans. **[32]**

Sol. $\sum\limits_{r=1}^{\infty} \frac{n}{\sqrt{n^4+r^4}} - \frac{2nr^2}{(n^2+r^2)\sqrt{n^4+r^4}}$

$\sum\limits_{r=1}^{\infty} \frac{\frac{1}{n}}{\sqrt{1+\left(\frac{r}{n}\right)^4}} - \frac{2\left(\frac{1}{n}\right)\left(\frac{r}{n}\right)^2}{\left(1+\left(\frac{r}{n}\right)^2\right)\sqrt{1+\left(\frac{r}{n}\right)^4}}$

$\Rightarrow \int_0^1 \frac{dx}{\sqrt{1+x^4}} - \frac{2x^2\,dx}{(1+x^2)\sqrt{1+x^4}}$

$\Rightarrow \int_0^1 \frac{1-x^2}{(1+x^2)\sqrt{1+x^4}}\,dx$

$\Rightarrow \int_0^1 \frac{\frac{1}{x^2}-1}{\left(x+\frac{1}{x}\right)\sqrt{x^2+\frac{1}{x^2}}}\,dx$

$\Rightarrow -\int_0^1 \frac{1-\frac{1}{x^2}}{\left(x+\frac{1}{x}\right)\sqrt{\left(x+\frac{1}{x}\right)^2-2}}\,dx$

$x + \frac{1}{x} = t \Rightarrow 1 - \frac{1}{x^2}\,dx = dt$

$$\Rightarrow -\int_{\infty}^{2} \frac{dt}{t\sqrt{t^2-2}}$$

$$\Rightarrow -\int_{\infty}^{2} \frac{t\,dt}{t^2\sqrt{t^2-2}}$$

take $t^2 - 2 = \alpha^2$

$t\,dt = \alpha\,d\,\alpha$

$$\Rightarrow -\int_{\infty}^{\sqrt{2}} \frac{\alpha d\alpha}{(\alpha^2+2)\alpha}$$

$$\Rightarrow -\int_{\infty}^{\sqrt{2}} \frac{d\alpha}{\alpha^2+2}$$

$$\Rightarrow \frac{-1}{\sqrt{2}}\tan^{-1}\frac{\alpha}{\sqrt{2}}\Big]_{\infty}^{\sqrt{2}}$$

$$\Rightarrow \frac{-1}{\sqrt{2}}\tan^{-1}1 + \frac{1}{\sqrt{2}}\tan^{-1}\infty$$

$$\Rightarrow \frac{1}{\sqrt{2}}\left\{\frac{\pi}{2} - \frac{\pi}{4}\right\}$$

$$\Rightarrow \frac{\pi}{4\sqrt{2}} = \frac{\pi}{K}$$

So, $K = 4\sqrt{2}$

$K^2 = 32$

Q.85 The remainder when 428^{2024} is divided by 21 is .....

Ans. **[1]**

Sol. $(428)^{2024} = (420 + 8)^{2024}$

$= (21 \times 20 + 8)^{2024}$

$= 21m + 8^{2024}$

Now, $8^{2024} = (8^2)^{1012}$

$= (64)^{1012}$

$= (63 + 1)^{1012}$

$= (21 \times 3 + 1)^{1012}$

$= 21n + 1$

$\Rightarrow$ Remainder is 1.

Q.86 Let $f : (0, \pi) \to R$ be a function given by

$$f(x) = \begin{cases} \left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}, & 0 < x < \frac{\pi}{2} \\ a - 8, & x = \frac{\pi}{2} \\ (1+|\cot|)^{\frac{b}{a}|\tan x|}, & \frac{\pi}{2} < x < \pi \end{cases}$$

where $a, b \in Z$. If f is continuous at $x = \frac{\pi}{2}$, then $a^2 + b^2$ is equal to .....

Ans. **[81]**

Sol. LHL at $x = \frac{\pi}{2}$

$$\lim_{x\to\frac{\pi}{2}}\left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}} = \left(\frac{8}{7}\right)^{0} = 1$$

RHL at $x = \frac{\pi}{2}$

$$\lim_{x\to\frac{\pi}{2}}(1+|\cot x|)^{\frac{b}{a}|\tan x|}$$

$$= e^{\lim\limits_{x\to\frac{\pi}{2}}|\cot x|\frac{b}{a}|\tan x|} = e^{\frac{b}{a}}$$

$\Rightarrow 1 = a - 8 = e^{b/a}$

$\Rightarrow a = 9, b = 0$

$\Rightarrow a^2 + b^2 = 81$

Q.87 Let A be a non-singular matrix of order 3. If $\det(3\text{adj}(2\text{adj}((\det A)A))) = 3^{-13}$, 2^{-10} and $\det(3\text{adj}(2A)) = 2^m \cdot 3^n$, then $|3m + 2n|$ is equal to .....

Ans. **[14]**

Sol. $|3\ \text{adj}(2\text{adj}(|A|A))| = |3\text{adj}(2|A|^2\ \text{adj}(A)|$

$= |3.2^2|A|^4\ \text{adj}(\text{adj}\ (A)| = 2^6 3^3 |A|^{12}|A|^4$

$= 2^6 3^3 |A|^{16} = 2^{-10}\ 3^{-13}$

$\Rightarrow |A|^{16} = 2^{-16}3^{-16} \Rightarrow |A| = 2^{-1}3^{-1}$

Now, $|3\text{adj}\ (2A)| = |3.2^2\ \text{adj}(A)|$

$= 2^6\ 3^3\ |A|^2 = 2^{-m}\ 3^{-n}$

$\Rightarrow 2^6\ 3^3\ 2^{-2}\ 3^{-2} = 2^{-m}\ 3^{-n}$

$\Rightarrow 2^{-m}\ 3^{-n} = 2^4\ 3^1$

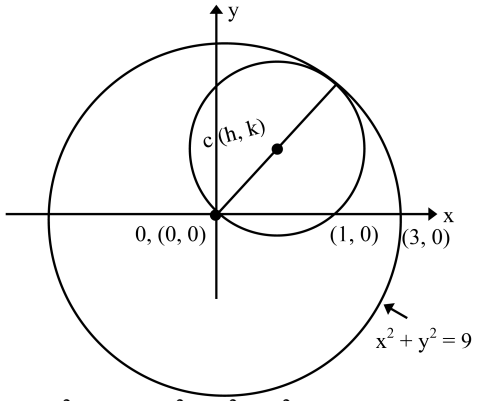
$\Rightarrow m = -4, n = -1$

$\Rightarrow |3m + 2n| = |-12 - 2| = 14$

Q.88 Let the centre of a circle, passing throguh the point (0, 0), (1, 0) and touching the circle $x^2 + y^2 = 9$, be (h, k). Then for all possible values of the coordinates of the centre (h, k), $4(h^2 + k^2)$ is equal to .....

Ans. **[9]**

Sol.



$(x - h)^2 + (y - k)^2 = h^2 + k^2$

$x^2 + y^2 - 2hx - 2ky = 0$

$\because$ passes through (1, 0)

$\Rightarrow 1 + 0 - 2h = 0$

$\Rightarrow h = ½$

$\therefore OC = \frac{OP}{2}$

$\sqrt{\left(\frac{1}{2}\right)^2 + k^2} = \frac{3}{2}$

$\frac{1}{4} + k^2 = \frac{9}{4}$

$k^2 = 2$

$k = \pm \sqrt{2}$

$\therefore$ Possible coordinate of

c(h, k) $\left(\frac{1}{2}, \sqrt{2}\right)\left(\frac{1}{2}, -\sqrt{2}\right)$

$4(h^2 + k^2) = 4\left(\frac{1}{4} + 2\right) = 4\left(\frac{9}{4}\right) = 9$

Q.89 If a function f satisfies $f(m + n) = f(m) + f(n)$ for all $m, n \in N$ and $f(1) = 1$, then the largest natural number λ such that $\sum_{k=1}^{2022} f(\lambda + k) \le (2022)^2$ is equal to .....

Ans. **[1010]**

Sol. $f(m + n) = f(m) + f(n)$

$\Rightarrow f(x) = kx$

$\Rightarrow f(1) = 1$

$\Rightarrow k = 1$

$f(x) = x$

Now, $\sum_{k=1}^{2022} f(\lambda + k) \le (2022)^2$

$\Rightarrow \sum_{k=1}^{2022} (\lambda + k) \le (2022)^2$

$\Rightarrow 2022\lambda + \frac{2022 \times 2023}{2} \le (2022)^2$

$\Rightarrow \lambda \le 2022 - \frac{2023}{2}$

$\Rightarrow \lambda \le 1010.5$

$\therefore$ largest natural no. λ is 1010

Q.90 Let A = {2, 3, 6, 7} and B = {4, 5, 6, 8}. Let R be a relation defined on A × B by (a_1, b_1) R (a_2, b_2) is and only if $a_1 + a_2 = b_1 + b_2$. Then the number of element in R is .....

Ans. **[25]**

Sol. A = {2, 3, 6, 7}

B = {2, 5, 6, 8}

(a_1, b_1) R(a_2, b_2)

$a_1 + a_2 = b_1 + b_2$

1. (2, 4) R (6, 4)	2. (2, 4) R (7, 5)	× 2
3. (2, 5) R (7, 4)	4. (3, 4) R (6, 5)	
5. (3, 5) R (6, 4)	6. (3, 5) R (7, 5)	
7. (3, 6) R (7, 4)	8. (3, 4) R (7, 6)	
9. (6, 5) R (7, 8)	10. (6, 8) R (7, 5)	
11. (7, 8) R (7, 6)	12. (6, 8) R (6, 4)	
13. (6, 6) R (6, 6)		

Total 24 + 1 = 25

CAREER POINT